

RDHO-Based Capacity-Feasible Joint Task Offloading and Computing Resource Allocation in Mobile Edge Computing: A Controlled Evaluation

Haoru Su*, Jie Bai, and Jiangyi Zhou

College of Computer Science, Beijing University of Technology, Beijing 100124, China

*Corresponding author: Haoru Su (suhaoru@bjut.edu.cn).

E-mail addresses: suhaoru@bjut.edu.cn (H. Su), baijie0219@gmail.com (J. Bai),
zhoujiangyi@bjut.edu.cn (J. Zhou).

Abstract

Mobile edge computing (MEC) complements resource-limited devices with nearby edge and remote cloud resources, but heterogeneous tasks couple discrete execution-node choice with continuous processor allocation. This paper develops an RDHO-based capacity-feasible framework for joint task offloading and computing resource allocation: legal-node encoding and physical CPU decoding are coupled with deterministic capacity repair, RDHO population search, fixed-objective incumbent tracking and optional coordinate refinement. Under the configured end-to-end procedures, RDHO-full achieves the lowest mean reporting fitness. However, strictly controlled experiments with common initialisation, a shared decoder and repair path, equal total evaluation budgets and, where applicable, common refinement show that RDHO significantly outperforms RIME but is significantly outperformed by DBO. The evidence therefore supports the RDHO-based integrated framework rather than universal superiority of the hybrid population operator. Conclusions are limited to the stated simulated model and implemented baselines.

Keywords: mobile edge computing; RDHO; joint task offloading; computing resource allocation; capacity feasibility; controlled evaluation.

1 Introduction

Latency-sensitive sensing, Internet of Things (IoT), fifth-generation (5G) and emerging sixth-generation services generate heterogeneous workloads close to users. Mobile edge computing (MEC) reduces service distance by supplementing devices with edge and cloud resources [1,2].

A practical offloading decision must select a legal execution node and allocate finite processor capacity. The resulting categorical-continuous decisions interact through transmission paths, per-node CPU budgets, device-side energy and task-specific service thresholds.

Existing studies primarily optimise energy and latency, while freshness-aware work adds Age of Information (AoI) and user-oriented work considers Quality of Experience (QoE) or fairness [3-7]. These criteria are related rather than statistically independent: in the present model AoI contains service delay and QoE transforms delay, energy and freshness into bounded utility.

To address this problem, this study develops an RDHO-based optimisation framework that adapts RIME- and DBO-inspired population movements to legal-node selection, physical CPU allocation and deterministic capacity repair. The framework treats the benefit of the hybrid population mechanism as an empirical question rather than an assumption.

The contributions are threefold. First, the paper formulates a capacity-feasible three-tier MEC model with legal local, edge and cloud execution, physical CPU-cycle-rate allocation, per-node aggregate capacity, device-side energy, delay, periodic AoI, priority-weighted aggregate QoE and priority-neutral active-user fairness. Second, it develops an RDHO-based integrated framework with legal-node population encoding, physical-resource decoding, deterministic reassignment and capacity projection, RIME-DBO-inspired population updates, dynamic search guidance, fixed reporting incumbents and optional coordinate refinement. Third, it reports paired controlled experiments with equal NFE, common initialisation, common repair, common refinement, Wilcoxon tests, Holm correction, rank-biserial effect sizes and wins/ties/losses. These experiments distinguish complete-pipeline evidence from isolated population-update evidence and show RDHO better than RIME but not DBO under the stated controls.

Under the configured end-to-end procedures, RDHO-full obtains mean reporting fitness 0.947. This configured comparison has unequal NFE and evaluates complete solvers, not an isolated RIME-DBO population operator. The strictly controlled results reported in Section 6 show the narrower population-stage and common-pipeline conclusions.

Section 2 reviews related work. Sections 3 and 4 define the physical system and P1. Section 5 presents the RDHO-based capacity-feasible optimisation framework. Section 6 separates configured end-to-end and controlled population-update evidence, and Section 7 concludes.

2 Related Work

Following MEC surveys, related work is organised by solution methodology: model-based optimisation, learning-based methods and metaheuristic search [8-10]. Objective dimensions are discussed within these classes.

Model-based studies jointly allocate communication and computation resources or coordinate edge-cloud execution under explicit system constraints [3,11-14]. Legal collaboration topology and resource budgets must be represented before optimising task placement.

Learning-based methods adapt policies to changing states and have been applied to online offloading, vehicular cooperation and blockchain-enabled edge-cloud systems [15-17]. Their training and data requirements motivate complementary transparent offline optimisation for a fixed decision epoch.

Freshness-aware MEC incorporates AoI alongside service time [18-20]. The earlier TLBO-HHO method is included as an AoI-aware metaheuristic baseline [21]. QoE and fairness studies represent service acceptability and user balance [5-7,22-26].

Metaheuristic methods search nonlinear mixed spaces without a learned policy. Relevant mechanisms include HHO, GA, TLBO, RIME, DBO and enhanced sparrow search [27-36]. A hybrid label alone does not establish benefit; initialisation, evaluation budget, population updates and refinement require controlled comparisons.

Representative studies on parking-edge cooperation, blockchain-enabled edge-cloud offloading, cooperative deep reinforcement learning and potential-game strategies show how collaboration topology, resource coupling and service requirements alter offloading decisions [37-40]. A remaining methodological gap is a reproducible framework that combines capacity-feasible physical resource allocation with priority-weighted system QoE and priority-neutral active-user fairness, while experimentally separating the effects of initialisation, evaluation budget, population updates and refinement.

The present work addresses that combined methodological and evaluation gap for a simulated cloud-edge-device epoch; it does not claim that physical CPU modelling alone is novel, nor does it claim communication-resource optimisation, online queue scheduling or deployment validation.

3 System Model

3.1 MEC Network, Assumptions, and Task Model

Consider devices D , edge servers E , cloud servers C and tasks T in a three-tier architecture. Their union $N = D \cup E \cup C$ uses global node IDs. We assume fixed topology and link rates during one decision epoch so that the optimisation focuses on execution-node choice and computing-resource allocation. Figure 1 illustrates the three-tier architecture, legal execution-node choices and shared physical CPU-capacity pools considered in one decision epoch.

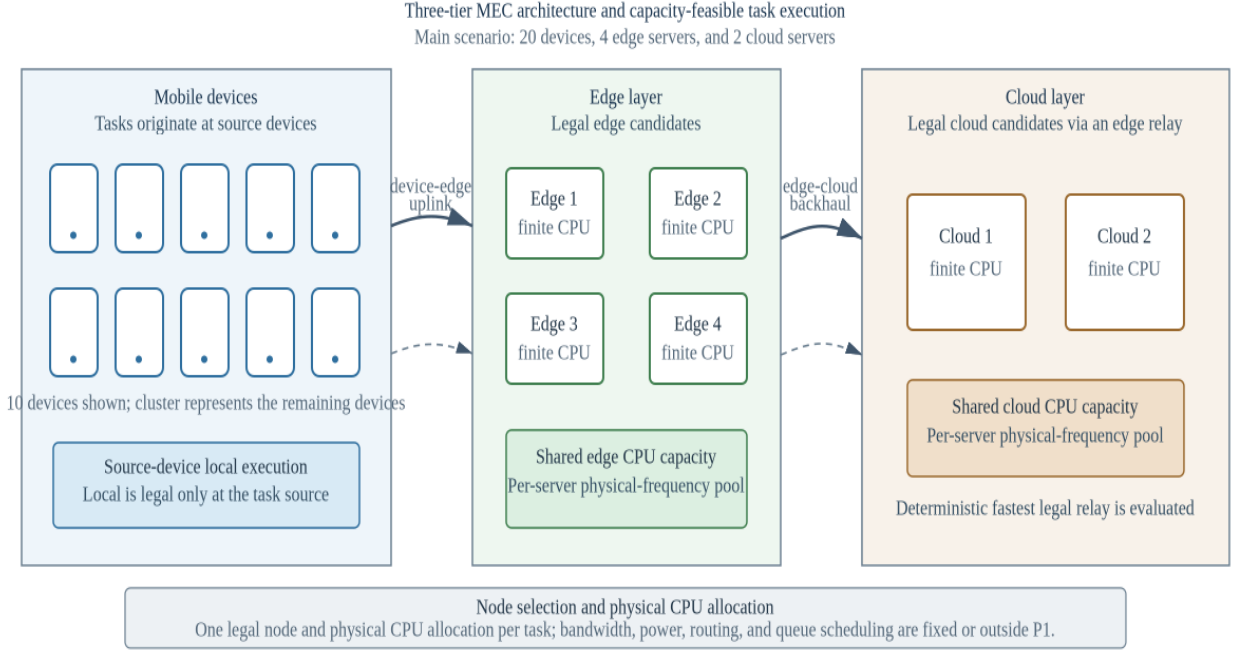


Fig. 1. Three-tier cloud-edge-device architecture with legal execution-node choices and shared physical CPU-capacity pools.

Task i is generated by source device $d(i)$. Local execution is legal only at $d(i)$; reachable edge nodes have positive device-edge rates; a cloud node is legal only through at least one reachable edge with a positive backhaul rate. For a selected cloud, evaluation uses the legal relay with minimum task-specific transmission delay. This is deterministic path evaluation, not route or bandwidth optimisation. Task i is represented by the following tuple, which records input bits, CPU cycles, period, service thresholds, battery ratio, priority and source device.

$$\tau_i = (b_i, c_i, \Delta_i, T_i^{\max}, E_i^{bud}, A_i^{\max}, \beta_i, \pi_i, d(i))$$

Binary $x_{ij} = 1$ selects exactly one node j . The derived layer z_i is local, edge or cloud and s_i denotes the selected remote server. The physical allocation f_i is measured in Hz.

Each search individual stores two normalised coordinates per task. The node coordinate indexes the sorted legal-node set, while $r_i \in [0, 1]$ decodes to a tentative physical frequency. The normalised coordinate is internal algorithm encoding, not a system-level abstract control variable.

3.2 Delay, Energy, and AoI Models

For node j with minimum allocatable frequency $f_j^{\min}$ and capacity F_j , the tentative request is

$$f_i^{req} = f_j^{\min} + r_i(F_j - f_j^{\min}), 0 \leq r_i \leq 1, x_{ij} = 1 \quad (1)$$

When reassignment is required, tasks are processed in a deterministic identifier order used solely for reproducibility. Identifiers are assigned independently of task type, source device and priority; the corresponding neutrality audit is provided in the reproducibility repository.

$$T_j = \{i \in T : x_{ij} = 1\}, n_j = \text{card}(T_j), R_j = F_j - n_j f_j^{\min} \quad (2)$$

Reassignment repeats for at most $|T||N|$ passes and either reaches non-negative R_j for every node or fails explicitly. For a feasible assignment, the projection below applies only to $i \in T_j$. A zero excess denominator necessarily selects the first branch, so no division is performed; feasible requests are retained and only overloaded excess is scaled.

$$\begin{aligned}
q_i &= \max\{f_i^{req} - f_j^{min}, 0\}, i \in T_j, \\
f_i &= f_j^{min} + \begin{cases} q_i, & \sum_{k \in T_j} q_k \leq R_j, \\ R_j q_i / \sum_{k \in T_j} q_k, & \sum_{k \in T_j} q_k > R_j. \end{cases}
\end{aligned} \tag{3}$$

Using fixed positive rates $R_{d,e}$ and $R_{e,c}$, end-to-end delay is defined separately for each execution layer.

$$T_i = \begin{cases} c_i / f_i, & z_i = local, \\ b_i / R_{d(i),s_i} + c_i / f_i + \delta_E, & z_i = edge, \\ b_i / R_{d(i),e_i} + b_i / R_{e_i,s_i} + c_i / f_i + \delta_C, & z_i = cloud, \end{cases} \tag{4}$$

No heuristic congestion attenuation or M/M/1 queue is added, because finite CPU capacity and repaired physical allocations already represent processor scarcity. Fixed edge/cloud overheads are stated simulation parameters.

For local execution, device-side dynamic-voltage-and-frequency-scaling energy is

$$E_i = \kappa_{d(i)} c_i f_i^2, z_i = local \tag{5}$$

For edge or cloud offloading, let e_i^* denote the selected edge or the fixed legal cloud relay. The reported device-side energy includes uplink transmission only:

$$E_i = P_{d(i)} \frac{b_i}{R_{d(i),e_i^*}}, z_i \in \{edge, cloud\} \tag{6}$$

Infrastructure computation and backhaul energy are outside the boundary; the manuscript therefore never labels this metric total system energy.

We assume periodic update generation and no explicit queue backlog. The area of a delay-shifted sawtooth over one period gives the average approximation

$$A_i = \frac{\Delta_i}{2} + T_i \tag{7}$$

AoI is consequently delay-coupled, not independent. The model is not a queue-aware or peak-AoI formulation.

Table 1 lists one core variable per row. Secondary symbols are defined at first use in the surrounding text to avoid an oversized notation table.

The generated ranges ensure a feasible minimum-frequency assignment; all returned solutions are checked for unique assignment, legal nodes, finite frequencies, bounds and aggregate capacity.

Table 1. Main Notation

Symbol	Definition
T	Task set
N	Unified device, edge and cloud execution-node set
N_i	Legal execution-node set of task i
$d(i)$	Source device of task i
x_{ij}	Binary assignment of task i to node j
r_i	Internal normalised CPU coordinate
f_i^{req}	Tentative physical CPU request of task i (Hz)
f_i	Repaired physical CPU allocation of task i (Hz)
f_i^{min}	Minimum allocatable CPU frequency of node j (Hz)
F_j	Total CPU capacity of node j (Hz)
T_j	Tasks assigned to node j after reassignment
n_j	Number of tasks assigned to node j
R_j	CPU capacity remaining after minimum allocations

Symbol	Definition
N_T	Number of tasks in the decision epoch
N_{act}	Number of active source devices
U_m	Mean base utility of active source device m
Q	Priority-weighted aggregate QoE
J	Jain fairness over active-user mean utilities
$B(X)$	Fixed weighted base objective
$CSR(X)$	Soft threshold constraint-satisfaction ratio
$F_{report}(X)$	Fixed problem objective and cross-algorithm scale
$F_{search}(X, t)$	Iteration-specific internal search fitness
$\lambda(t)$	Dynamic search penalty coefficient
NFE	Number of function evaluations

141 3.3 QoE and Fairness Metrics

142 The study uses a model-based base utility u_i rather than a human-subject mean-opinion score. Delay,
143 energy and freshness satisfaction are bounded exponential functions.

144 Delay satisfaction is

$$145 \quad S_i^T = \exp\left(-\frac{T_i}{T_i^{max}}\right) \quad (8)$$

146 Energy satisfaction is

$$147 \quad S_i^E = \exp\left(-\frac{E_i}{E_i^{bud}}\right) \quad (9)$$

148 Freshness satisfaction is

$$149 \quad S_i^A = \exp\left(-\frac{A_i}{A_i^{max}}\right) \quad (10)$$

150 The priority-neutral base task utility is

$$151 \quad u_i = 0.45 S_i^T + 0.30 S_i^E + 0.25 S_i^A, 0 < u_i \leq 1 \quad (11)$$

152 Task priority is used only for system QoE aggregation:

$$153 \quad Q = \frac{\sum_{i \in T} \pi_i u_i}{\sum_{i \in T} \pi_i} \quad (12)$$

154 For each active source device m , let U_m denote the mean base utility over tasks generated by m .

155 This ordering prevents task priority and the number of generated tasks from being counted again
156 inside fairness.

157 Let D_{act} contain only source devices that generate at least one task in the epoch, and let N_{act} be their
158 count.

$$159 \quad J = \frac{\left(\sum_{m \in D_{act}} U_m\right)^2}{N_{act} \sum_{m \in D_{act}} U_m^2} \quad (13)$$

160 $J = 1$ indicates equal active-user mean base utility. It is QoE-outcome fairness, not equality of physical
161 CPU allocations.

162 The utility coefficients are engineering parameters applied identically to all algorithms; they are not
 163 claimed as subjective calibration and are varied in sensitivity analysis.

164 4 Problem Formulation

165 P1 combines threshold-normalised device-side energy, delay and AoI with QoE and fairness deficits.
 166 These five terms are coupled, because QoE is derived from the first three and AoI contains delay.

167 The components E^{norm} , D^{norm} and A^{norm} are respectively the task means of device-side energy divided
 168 by its budget, delay divided by its maximum and AoI divided by its threshold.

169 The fixed base objective is

$$170 \quad B(X) = 0.15 E^{norm} + 0.15 D^{norm} + 0.20 A^{norm} + 0.25(1 - Q) + 0.25(1 - J) \quad (14)$$

171 Let N_T be the number of tasks. Soft delay, battery-adjusted energy and AoI checks are summarised by
 172 CSR; they do not replace the hard assignment and CPU constraints.

$$173 \quad CSR(X) = \frac{1}{3N_T} \sum_{i \in T} [1(T_i \leq T_i^{max}) + 1(E_i \leq \max\{\beta_i, 0.1\} E_i^{bud}) + 1(A_i \leq A_i^{max})] \quad (15)$$

174 The fixed optimisation problem is therefore defined directly on reporting fitness. The selected-node
 175 CPU bound is written as an assignment-weighted bound, so each f_i is constrained only by its chosen
 176 node.

$$\begin{aligned} 177 \quad P1: \min_X F_{report}(X) &= B(X) + \lambda_{ref}[1 - CSR(X)] \\ &s.t. \sum_{j \in N} x_{ij} = 1, x_{ij} \in \{0, 1\}, i \in T, \\ &\quad x_{ij} = 0, j \notin N_i, \\ &\quad \sum_{j \in N} x_{ij} f_j^{min} \leq f_i \leq \sum_{j \in N} x_{ij} F_j, i \in T, \\ &\quad \sum_{i \in T} x_{ij} f_i \leq F_j, j \in N, \lambda_{ref} = 1. \end{aligned} \quad (16)$$

178 The soft-violation rate is $v = 1 - CSR$.

179 The internal normalised coordinates satisfy $0 \leq r_i \leq 1$ and select only the legal list; they are not
 180 additional physical constraints.

181 The dynamic search fitness used by RDHO is

$$182 \quad F_{search}(X, t) = B(X) + \lambda(t)[1 - CSR(X)] \quad (17)$$

$$183 \quad \lambda(t) = \lambda_0 \left(1 + \frac{2t}{T_{max}}\right)^\alpha \quad (18)$$

184 Parents and candidates in one greedy selection are evaluated with the same $\lambda(t)$. This prevents
 185 comparison under incompatible penalty coefficients.

186 The search process separately tracks the best fixed-objective candidate seen, denoted by C_{seen} :

$$187 \quad X_{report}^* = \arg \min_{X \in C_{seen}} F_{report}(X) \quad (19)$$

188 Search fitness guides optimisation; reporting fitness is the only cross-algorithm scale in tables and
 189 statistics.

190 Thus P1 is one fixed constrained problem, F_{search} is only a penalty-continuation mechanism for
 191 solving it, and F_{report} is the sole ranking scale in tables and statistics.

192

193 Problem complexity.

194 Even without continuous allocation, legal node assignments grow combinatorially. Capacity coupling,
 195 exponential utility and indicator-based soft violations make P1 a nonlinear mixed discrete-continuous
 196 problem.

197 The study does not claim a new proof of NP-hardness; it motivates a population search and evaluates
 198 empirical solution quality and cost.
 199 A returned incumbent consists of one repaired legal node and one physical CPU allocation per task.
 200 These characteristics motivate population search over legal categorical assignments and bounded
 201 physical CPU allocations; no deployment claim is made.

202 **5 RDHO-Based Capacity-Feasible Optimisation Framework**

203 RIME-DBO hybrid optimisation (RDHO) is the focal solver in this study. It adapts RIME-inspired
 204 perturbation and DBO-inspired role-conditioned movements to the capacity-feasible MEC encoding,
 205 decoder and repair path. RDHO-full additionally uses configured seeding and deterministic coordinate
 206 refinement; the framework is evaluated both as a complete solver and under controls that isolate its
 207 population-update stage.

208 RIME-inspired and DBO-inspired movements are described as algorithmic components, not as
 209 mechanisms presumed to be stronger in advance. Their independent contribution is evaluated under
 210 common initialisation, common repair, equal total NFE and common refinement in Section 6.3.

211 All candidates use the same normalised two-coordinate encoding and are decoded through the
 212 common physical repair before evaluation.

213 RDHO is assessed as a configured capacity-feasible search framework. Controlled experiments, rather
 214 than the hybrid label alone, determine which population-update claims are supported.

215 *5.1 Solution Encoding and Physical-Resource Decoding*

216 Each individual is a $|T| \times 2$ matrix (node coordinate, resource coordinate). The first coordinate indexes
 217 the sorted legal nodes; the second decodes to f_i^{req} in Eq. (1). Both are clipped to $[0,1]$.

218 Server selection is therefore an implemented decision with heterogeneous legal paths. The
 219 deterministic repair in Eqs. (2)-(3) is shared by RDHO and every baseline.

220 A candidate evaluation returns $B(X)$, $CSR(X)$, dynamic search fitness, fixed reporting fitness, raw
 221 metrics, hard-feasibility flags and capacity utilisation.

222 *5.2 RDHO Population-Update Mechanism*

223 The RDHO population contains equal Gaussian and uniform subsets, a greedy coordinate seed and up
 224 to three perturbations. Role shares adapt to diversity, while the configured top 10% are retained as
 225 elites.

226 Producer candidates combine RIME- and DBO-inspired updates through a weight decreasing from 0.8
 227 to 0.2; followers use best-guided puncture or bound-aware foraging, and scouts use theft or a decaying
 228 Cauchy perturbation.

229 The fusion weight is

$$230 \quad \omega(t) = 0.8 - 0.6 \frac{t}{T_{max}} \quad (20)$$

$$231 \quad X_k^{new} = \omega(t) X_k^{RIME} + [1 - \omega(t)] X_k^{DBO} \quad (21)$$

232 The continuous update is an algorithmic neighbourhood only; every resulting assignment is interpreted
 233 through the task's legal-node list.

234 Greedy replacement compares parent and candidate under the same current $\lambda(t)$. A separate
 235 incumbent is tracked on F_{report} .

236 RDHO-full applies deterministic coordinate refinement and then reports the common fixed objective.
 237 In the controlled experiments, RDHO-specific local refinement is disabled and the same deterministic
 238 refinement is either omitted for every method or applied to every method. This separates complete-
 239 pipeline evidence from the population-update comparison.

240 The complete procedure is summarised in Algorithm 1.

241 Algorithm 1. RDHO-based capacity-feasible joint task-offloading and physical CPU-allocation
 242 strategy.
 243

Algorithm 1. RDHO-based capacity-feasible joint offloading and physical CPU allocation	
Require:	Scenario, legal-node encoding, physical decoder, deterministic repair, objective weights, population size and iterations
Ensure:	Hard-feasible incumbent on the fixed reporting objective
1	Generate the configured RDHO population and optional seed candidates
2	Decode legal nodes; repair capacity; evaluate the base objective and soft CSR
3	for each configured iteration do
4	Evaluate parents with the current dynamic penalty and assign RDHO roles
5	Generate RIME-DBO-inspired producer, follower and scout candidates
6	Decode and deterministically repair every candidate
7	Compare parent and candidate under the same current penalty
8	Update the independent fixed-objective incumbent
9	end for
10	If configured, apply deterministic coordinate refinement
11	Re-evaluate with fixed reporting coefficient 1 and verify hard feasibility
Return	Node assignments, CPU frequencies and all reported metrics
Note	Controlled experiments replace RDHO-specific initialisation/refinement with the common procedure before comparing population updates

244 Algorithm 1 retains the RDHO population-update sequence while making its common decoder,
 245 deterministic capacity repair, fixed-reference incumbent tracking and optional refinement explicit.

246 6 Performance Evaluation

247 6.1 Experimental Setup and Reproducibility

248 The configured V2 end-to-end experiments and the additive strictly controlled experiments use the
 249 same fixed synthetic scenarios 20260701-20260730, legal-node encoding, physical CPU decoder,
 250 deterministic capacity repair, hard-feasibility checks and fixed reporting objective. The controlled
 251 artifacts are generated from immutable raw CSV files and audited separately; they do not regenerate or
 252 modify results/v2/.

253 The configured end-to-end suite compares RDHO-full, RIME, DBO, TLBO-HHO [21], CWTSSA
 254 [36] and Greedy-ED under the same model, utility, repair and fixed reporting objective.

255 The configured end-to-end comparison intentionally retains each solver's configured procedure; NFE
 256 is therefore unequal (RDHO-full 10,232, population baselines 7,551 and Greedy-ED 681). It is
 257 reported as a complete-solver comparison and cannot isolate the contribution of RDHO's population-
 258 update mechanism.

259 Table 2. Physical system parameters used in the main experiment.

Parameter	Value and interpretation
Devices / edge / cloud	20 / 4 / 2
Tasks	40 heterogeneous tasks
Node CPU capacity	Device 2.2-3.0; edge 18-28; cloud 55-75 GHz
Minimum allocatable CPU	Device 0.2; edge 0.8; cloud 1.5 GHz

Parameter	Value and interpretation
Transmit power	0.2-0.8 W
Device-edge rate	8-30 Mbit/s; 10% sparse links, connectivity repaired
Edge-cloud rate	60-150 Mbit/s; 5% sparse links, connectivity repaired
Energy coefficient	$(0.8-1.4) \times 10^{-27} \text{ J s}^2 \text{ cycle}^{-3}$
Fixed overhead	Edge 0.010 s; cloud 0.055 s
Cloud relay	Fastest legal fixed relay for each selected cloud
Excluded controls	Bandwidth, power, association, routing and queue scheduling

Table 3. Task-generation ranges and sampling probabilities.

Parameter	Compute-intensive	Data-intensive	Real-time	Lightweight
Sampling probability	0.28	0.24	0.28	0.20
Input data (MB)	8-35	30-90	2-15	0.5-5
CPU cycles (Gcycles)	1.8-4.5	0.8-2.2	0.5-1.8	0.1-0.8
Maximum delay (s)	1.5-3.0	2.5-5.0	0.35-1.0	0.8-2.0
AoI threshold (s)	2.0-3.0	3.0-5.0	0.5-1.0	1.0-2.0
Energy budget (J)	2.5-6.0	2.0-5.0	1.0-3.5	0.5-2.0
Battery ratio	0.45-1.0	0.40-0.95	0.55-1.0	0.50-1.0
Priority / interval (s)	0.60-0.90 / 0.50-1.00	0.50-0.80 / 0.80-1.50	0.80-1.00 / 0.10-0.30	0.40-0.70 / 0.30-0.60

Table 4. Algorithm and reproducibility parameters for the configured end-to-end comparison and controlled experiments.

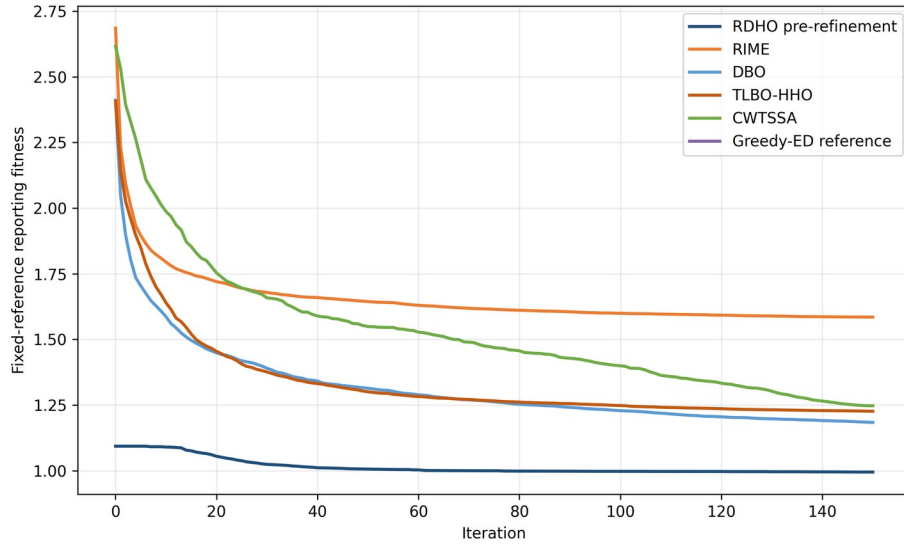
Parameter	Value
Configured end-to-end population / iterations	50 / 150
Objective weights (energy, delay, AoI, QoE, fairness)	0.15, 0.15, 0.20, 0.25, 0.25
Reporting coefficient	Fixed at 1
Paired scenarios	30 (seeds 20260701-20260730)
Experiment A control	Common initial population, decoder and repair; no refinement; 3,801 total NFE
Experiment B control	Common initial population, decoder, repair and deterministic refinement; 10,232 total NFE

Parameter	Value
Controlled statistics	Two-sided paired Wilcoxon, experiment-local Holm correction, rank-biserial effect size, W/T/L
Compared population methods	RDHO, RIME and DBO

265 No algorithm-specific tuning was performed on the reported scenarios. Configurations and exact
266 baseline implementations are versioned with the raw results.

267 6.2 Configured End-to-End Solver Comparison

268 Figure 2 reports fixed-reference incumbents during population search. RDHO's final coordinate
269 refinement is not included in its curve, and Greedy-ED is a fixed reference.



270

271

Fig. 2. Fixed-reference reporting-fitness convergence over 150 iterations.

272

Table 5. Configured end-to-end solver comparison over 30 paired scenarios (mean +/- standard deviation; unequal NFE).

273

Alg.	Reporting fitness	Device energy (J)	Delay (s)	AoI (s)	QoE	Fairness	Soft CSR	Time (s)	NFE
RDHO -full	0.947 (0.120)	94.58 (24.04)	1.848 (0.218)	2.171 (0.226)	0.447 (0.032)	0.924 (0.020)	0.721 (0.055)	7.831 (0.660)	10232
RIME	1.585 (0.154)	126.37 (30.12)	4.131 (0.795)	4.454 (0.813)	0.355 (0.027)	0.838 (0.040)	0.513 (0.053)	5.966 (0.499)	7551
DBO	1.184 (0.190)	93.19 (31.46)	2.536 (0.479)	2.859 (0.487)	0.415 (0.035)	0.916 (0.026)	0.629 (0.070)	5.820 (0.501)	7551
TLBO- HHO	1.226 (0.201)	60.63 (14.07)	2.766 (0.589)	3.089 (0.593)	0.407 (0.040)	0.910 (0.029)	0.599 (0.072)	5.711 (0.495)	7551
CWTS SA	1.247 (0.151)	62.62 (17.65)	2.775 (0.488)	3.098 (0.496)	0.408 (0.031)	0.914 (0.024)	0.588 (0.057)	5.678 (0.486)	7551
Greedy -ED	1.093 (0.170)	99.75 (22.47)	2.225 (0.538)	2.548 (0.550)	0.428 (0.034)	0.916 (0.032)	0.648 (0.062)	0.507 (0.044)	681

274

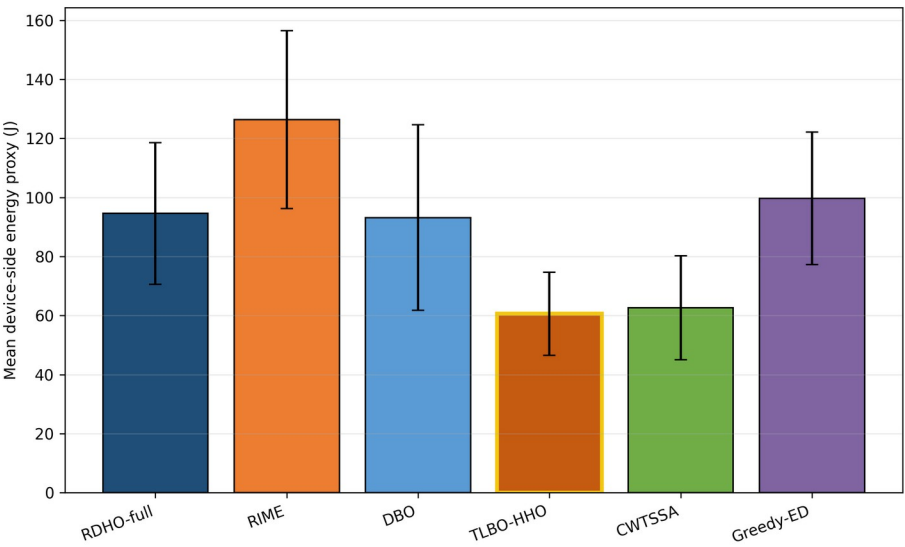
275

276

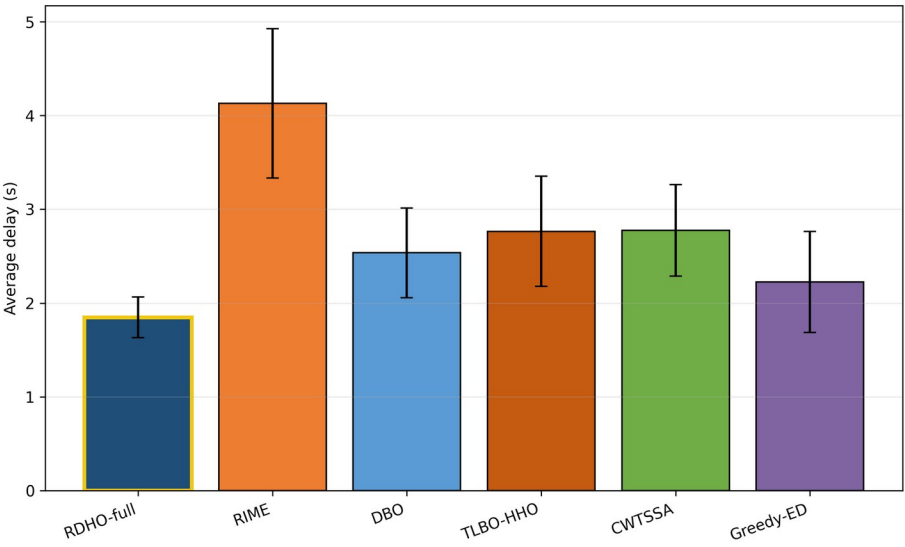
277

RDHO-full has the lowest configured mean reporting fitness (0.947 +/- 0.120) and every returned solution is hard feasible. Because RDHO-full uses 10,232 NFE while the population baselines use 7,551, this result supports the configured end-to-end solver rather than isolated superiority of its hybrid population update.

278 Metric-specific leaders differ: TLBO-HHO has the lowest device-side energy, RDHO-full has the
 279 lowest delay and AoI and the highest QoE, active-user fairness and CSR, while Greedy-ED is fastest.
 280 RDHO-full also consumes more NFE.



281
 282 Fig. 3. Mean device-side energy over 30 paired runs.
 283 TLBO-HHO is the energy leader; RDHO optimises energy as one component of the coupled reporting
 284 preference.



285
 286 Fig. 4. Mean processing delay over 30 paired runs.
 287 RDHO-full has the lowest mean delay under the configured end-to-end procedures.

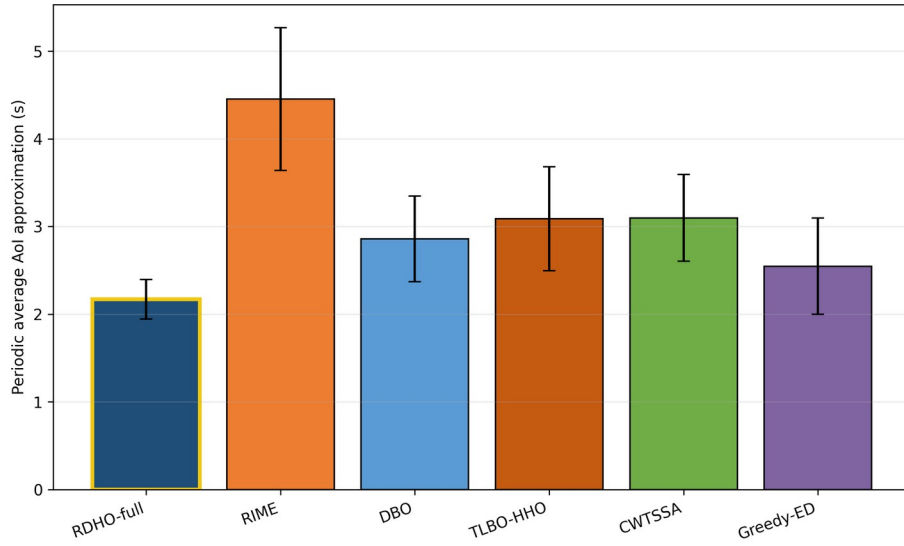


Fig. 5. Periodic no-backlog average-AoI approximation over 30 paired runs.

RDHO-full has the lowest mean AoI approximation; the close ordering with delay follows Eq. (7).

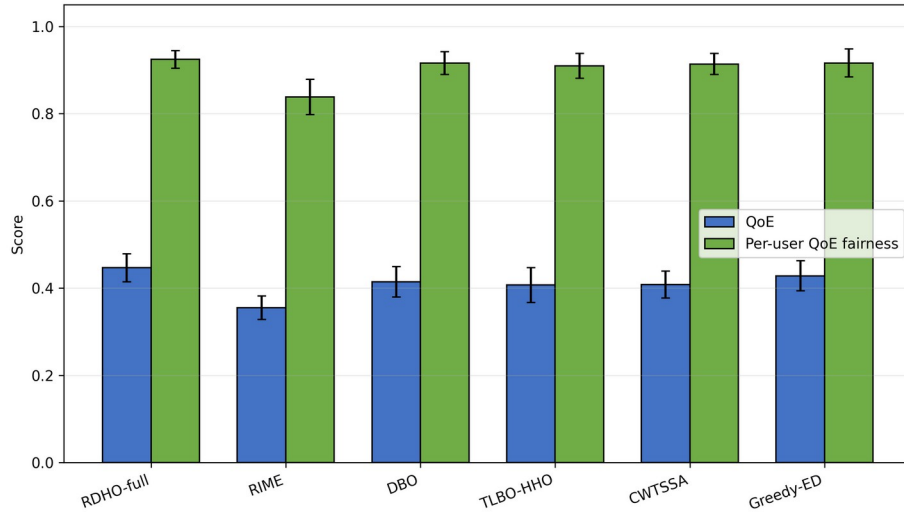


Fig. 6. Priority-weighted aggregate QoE and active-user base-utility fairness.

RDHO-full has the highest mean QoE, whereas fairness differences are small and do not imply equality of CPU allocations.

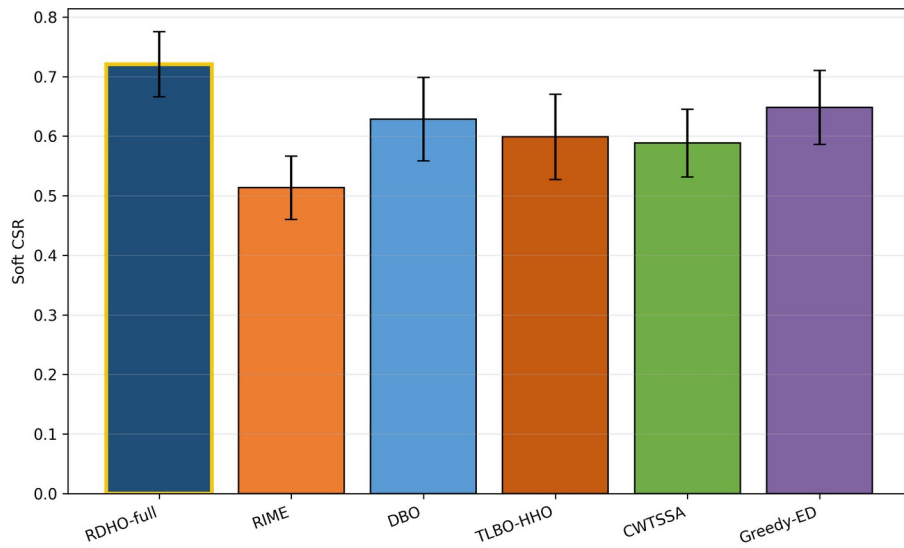


Fig. 7. Soft constraint-satisfaction ratio (CSR).

RDHO-full reaches mean CSR 0.721. CSR remains a binary threshold diagnostic; it does not imply that every task satisfies every soft threshold.

Configured-comparison statistics.

Two-sided paired Wilcoxon tests use the 30 matched reporting-fitness values. Holm adjustment controls the family-wise error rate; median paired difference, signed rank-biserial correlation and wins/ties/losses report magnitude and consistency. In this subsection, they test complete configured procedures with unequal NFE.

Table 6. Paired Wilcoxon tests for configured end-to-end solvers (unequal NFE).

Comparison	W	Two-sided p	Holm p	Median diff.	Signed r_{rb}	W/T/L
RDHO-full vs RIME	0	1.86e-09	9.31e-09	-0.6454	-1.000	30/0/0
RDHO-full vs DBO	0	1.86e-09	9.31e-09	-0.2030	-1.000	30/0/0
RDHO-full vs TLBO-HHO	0	1.86e-09	9.31e-09	-0.2581	-1.000	30/0/0
RDHO-full vs CWTSSA	0	1.86e-09	9.31e-09	-0.2979	-1.000	30/0/0
RDHO-full vs Greedy-ED	0	1.86e-09	9.31e-09	-0.1029	-1.000	30/0/0

RDHO-full is lower in all 30 pairs against each configured main baseline. These tests concern complete procedures with their configured initialisation, population search, incumbent tracking and refinement; they do not prove an isolated hybrid-operator advantage.

6.3 Controlled Evaluation of the RDHO Population Mechanism

Two strictly controlled paired experiments answer the attribution question directly. Every method receives one identical 50-by-40-by-2 initial population per scenario, follows the common legal-node decoder and deterministic repair path, reports the same fixed reporting fitness and is constrained by an exact total NFE budget. All 90 returns in each experiment are hard feasible. The primary endpoint is lower fixed reporting fitness; paired differences are defined as $\Delta F = F_{RDHO} - F_{parent}$, so negative values favour RDHO.

Table 7. Strictly controlled RDHO population-stage and common-pipeline evidence over 30 paired scenarios.

Experiment	Method	Mean +/- SD	Median	NFE	Runtime (s)
A: population stage	RDHO	1.4188 +/- 0.1612	1.3946	3801	3.149
A: population stage	RIME	1.6834 +/- 0.1651	1.6593	3801	3.154
A: population stage	DBO	1.2409 +/- 0.1245	1.2208	3801	3.053
B: common pipeline	RDHO	0.9732 +/- 0.1244	0.9827	10232	9.054
B: common pipeline	RIME	0.9836 +/- 0.1205	0.9950	10232	9.169
B: common pipeline	DBO	0.9664 +/- 0.1234	0.9676	10232	8.930

Experiment	Method	Mean +/- SD	Median	NFE	Runtime (s)
Paired tests	RDHO comparison	Median DeltaF [95% CI]	Holm p	r_rb	W/T/L
A	RDHO vs RIME	-0.278300 [-0.312822, -0.198123]	1.304e-08	-0.983	29/0/1
A	RDHO vs DBO	+0.178910 [+0.112584, +0.225303]	3.725e-09	+1.000	0/0/30
B	RDHO vs RIME	-0.012281 [-0.024215, -0.001839]	0.019863	-0.531	23/0/7
B	RDHO vs DBO	+0.008226 [+0.000775, +0.014349]	0.026229	+0.462	10/0/20

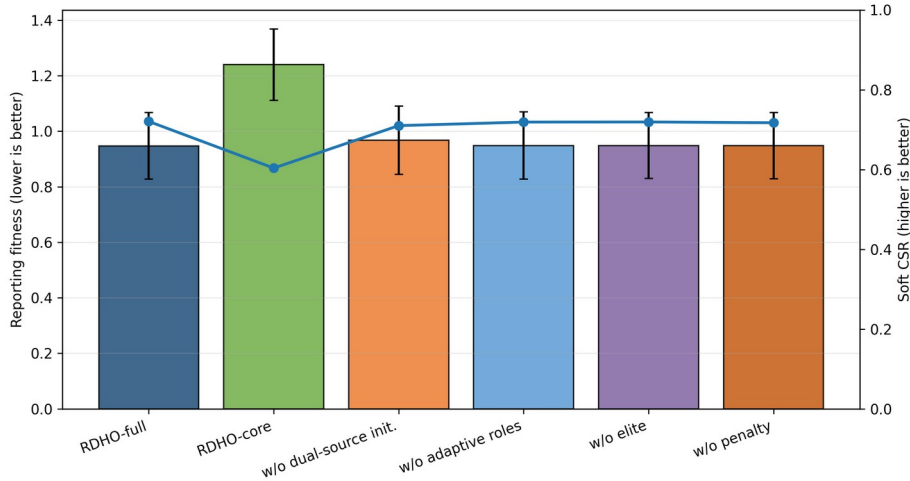


Fig. 8. Reporting fitness and soft CSR for configured RDHO variants.

The configured ablation remains descriptive for the end-to-end RDHO workflow. Coordinate refinement changes configured mean fitness from 1.240 to 0.947, so it is one component of the complete-pipeline result; the controlled experiments below are the evidence for population-update attribution.

6.4 Component, Refinement, Scalability, and Sensitivity Analyses

Equal-NFE population-stage control (Experiment A) uses common initialisation, decoder and repair, disables coordinate refinement, and fixes total NFE at 3,801 for every method. RDHO reports 1.4188 +/- 0.1612, RIME 1.6834 +/- 0.1651 and DBO 1.2409 +/- 0.1245. RDHO versus RIME has median DeltaF -0.278300, Holm p=1.304e-08, rank-biserial -0.983 and W/T/L 29/0/1; RDHO versus DBO has median DeltaF +0.178910, Holm p=3.725e-09, rank-biserial +1.000 and W/T/L 0/0/30. Thus RDHO significantly improves on RIME but is significantly worse than DBO.

Table 8. RDHO scalability under 20-100 tasks.

Tasks	Reporting fitness	Soft CSR	Time (s)	NFE
20	0.897 (0.124)	0.745 (0.059)	3.430 (0.056)	8892
40	0.977 (0.116)	0.708 (0.055)	6.641 (0.119)	10232
60	0.998 (0.106)	0.698 (0.050)	10.370 (0.128)	11572
80	1.059 (0.100)	0.665 (0.041)	14.833 (0.226)	12912

Tasks	Reporting fitness	Soft CSR	Time (s)	NFE
100	1.135 (0.070)	0.633 (0.030)	19.772 (0.220)	14252

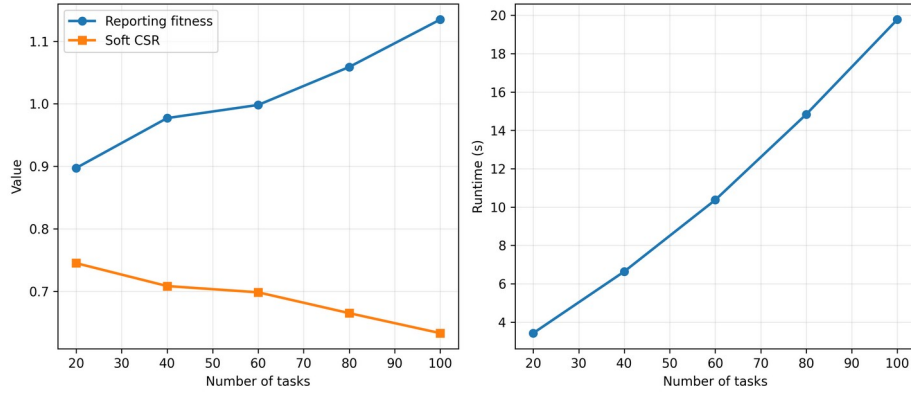


Fig. 9. Reporting fitness, soft CSR and runtime versus task count.

Common-pipeline control (Experiment B) additionally applies the same deterministic coordinate refinement to every method and fixes total NFE at 10,232. RDHO reports 0.9732 $\pm$ 0.1244, RIME 0.9836 $\pm$ 0.1205 and DBO 0.9664 $\pm$ 0.1234. RDHO versus RIME has median DeltaF -0.012281, Holm p=0.019863, rank-biserial -0.531 and W/T/L 23/0/7; RDHO versus DBO has median DeltaF +0.008226, Holm p=0.026229, rank-biserial +0.462 and W/T/L 10/0/20. RDHO again significantly improves on RIME but is significantly worse than DBO.

Component and refinement interpretation.

The controlled results indicate that the implemented RDHO population mechanism improves substantially over RIME but does not improve on DBO in the fixed V2 scenarios. The contribution of RDHO in this study is therefore interpreted at the integrated-framework level, not as evidence of a universally superior hybrid population operator. Common coordinate refinement reduces the absolute gaps, but it does not reverse the controlled RDHO-versus-DBO ordering.

The final sensitivity ranges are generated directly from V2 CSV files; no value is manually entered into a paper table.

Weight changes expose engineering preference rather than a universal optimum. Task-utility coefficient sensitivity checks the internal QoE construction, while CPU-capacity, SLA and server-heterogeneity scaling test load, threshold strictness and heterogeneous legal server choices.

Table 9. Additional configured attribution and sensitivity records (not a substitute for the strict controls in Table 7).

Method	Equal NFE	Common init.	Common refine.	Reporting fitness, mean (SD)	Effect size (reference - row)
RDHO-core	Yes (3,801)	No	No	1.402 (0.165)	Reference
RIME	Yes (3,801)	No	No	1.633 (0.161)	-0.978
DBO	Yes (3,801)	No	No	1.246 (0.145)	1.000
TLBO-HHO	Yes (3,801)	No	No	1.240 (0.139)	0.996
CWTSSA	Yes (3,801)	No	No	1.330 (0.158)	0.733

Method	Equal NFE	Common init.	Common refine.	Reporting fitness, mean (SD)	Effect size (reference - row)
RIME + refine	No	Yes	Yes	0.982 (0.129)	-1.000
DBO + refine	No	Yes	Yes	0.967 (0.127)	-0.892
RDHO-full	No (10,232)	RDHO	Yes	0.947 (0.120)	Reference

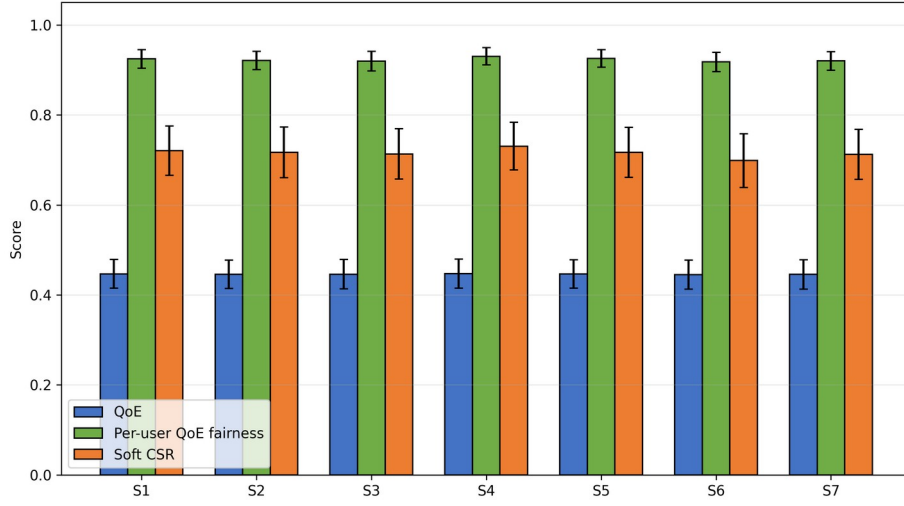


Fig. 10. Objective-weight sensitivity of QoE, active-user fairness and soft CSR.

The penalty heatmaps compare returned solutions under a common $\lambda_{ref}=1$; they do not alter the reporting definition.

Table 10. Dynamic-penalty sensitivity under fixed reporting fitness.

Initial penalty	Growth exponent	Reporting fitness	QoE	Fairness	Soft CSR	Time (s)
0.5	1.0	0.947 (0.119)	0.447 (0.032)	0.924 (0.020)	0.719 (0.057)	6.661 (0.113)
0.5	2.0	0.948 (0.118)	0.447 (0.032)	0.924 (0.020)	0.719 (0.055)	6.606 (0.106)
0.5	3.0	0.948 (0.119)	0.447 (0.033)	0.925 (0.020)	0.720 (0.055)	6.519 (0.109)
1.0	1.0	0.949 (0.119)	0.447 (0.033)	0.924 (0.020)	0.719 (0.054)	6.521 (0.084)
1.0	2.0	0.947 (0.120)	0.447 (0.032)	0.924 (0.020)	0.721 (0.055)	6.514 (0.113)
1.0	3.0	0.948 (0.121)	0.446 (0.033)	0.924 (0.020)	0.721 (0.054)	6.536 (0.113)
2.0	1.0	0.948 (0.120)	0.447 (0.032)	0.925 (0.020)	0.721 (0.055)	6.532 (0.106)
2.0	2.0	0.948 (0.119)	0.447 (0.032)	0.925 (0.020)	0.721 (0.053)	6.534 (0.118)

Initial penalty	Growth exponent	Reporting fitness	QoE	Fairness	Soft CSR	Time (s)
2.0	3.0	0.950 (0.121)	0.446 (0.033)	0.925 (0.020)	0.722 (0.054)	6.520 (0.109)

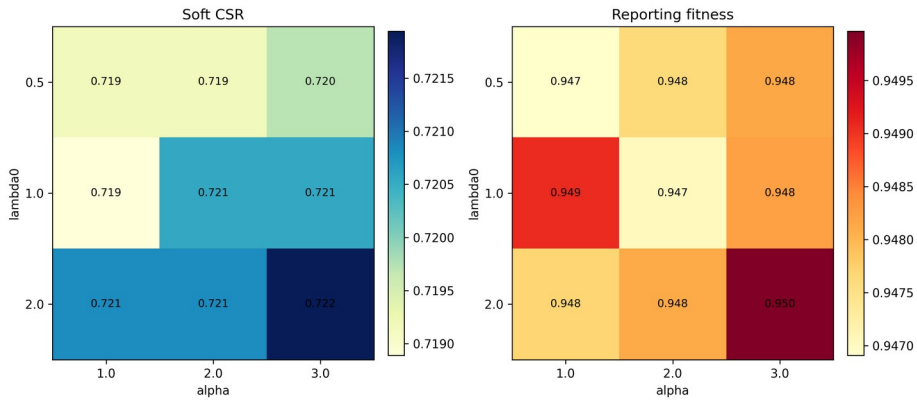
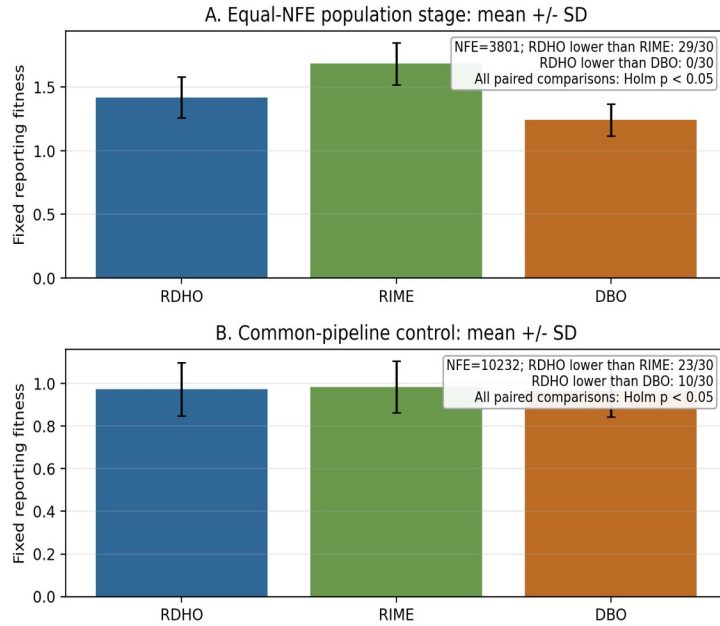


Fig. 11. Dynamic-penalty sensitivity heatmaps for soft CSR and reporting fitness.

The nine penalty schedules are reported without post-hoc selection. Full objective-weight, task-utility, CPU-capacity, SLA and server-heterogeneity tables remain in the repository supplement and extend the sensitivity boundary without post-hoc tuning claims.

Scalability and sensitivity.

Strictly controlled RDHO evidence (lower reporting fitness is better)



Overall interpretation.

Figure 12. Strictly controlled paired reporting-fitness evidence. Panels A and B use 3,801 and 10,232 total NFE, respectively; negative RDHO-minus-parent differences favour RDHO.

Figure 12 visualises both controlled experiments rather than hiding the adverse comparison: RDHO is lower than RIME in 29/30 population-stage pairs and 23/30 common-pipeline pairs, but it is lower than DBO in 0/30 and 10/30 pairs, respectively. The Holm-adjusted tests are significant in all four comparisons. The configured RDHO-full result therefore demonstrates the effectiveness of the complete RDHO-based framework under its configured procedure, whereas the isolated population-update evidence does not support superiority over DBO.

374 **7 Conclusion**

375 This work formulated a capacity-feasible MEC joint task-offloading and physical CPU-allocation
376 model and developed RDHO as its focal integrated solver framework. Each task selects one legal
377 source-local, reachable-edge or reachable-cloud node, and deterministic repair enforces physical Hz
378 bounds and per-node capacity.

379 In the configured end-to-end comparison, RDHO-full has the lowest mean fixed reporting fitness
380 (0.947) with hard feasibility in every returned solution, but this comparison has unequal NFE and
381 evaluates complete solver configurations. The strict controls disclose the narrower result: at 3,801
382 NFE without refinement, RDHO is significantly lower than RIME but significantly higher than DBO;
383 at 10,232 NFE with common refinement, RDHO is again significantly lower than RIME but
384 significantly higher than DBO. Accordingly, the evidence supports the RDHO-based capacity-feasible
385 framework and its controlled evaluation methodology, not independent or universal superiority of the
386 RIME-DBO population update.

387 Limitations include synthetic offline tasks, fixed-rate communication, deterministic cloud relay
388 selection, device-side rather than infrastructure energy, a periodic no-backlog AoI approximation,
389 coupled objective terms, engineering utility coefficients and the fixed set of implemented algorithms
390 and 30 scenarios. Future work can investigate adaptive operator selection or DBO-dominant RDHO
391 variants, queue-aware arrivals, communication-resource optimisation, infrastructure energy, calibrated
392 QoE and physical testbeds without claiming that the present hybrid operator has already surpassed
393 DBO.

394 **CRedit authorship contribution statement**

395 Haoru Su: Conceptualization, methodology, supervision, Writing - review and editing. Jie Bai:
396 Methodology, validation, formal analysis, Writing - review and editing. Jiangyi Zhou: Software, data
397 curation, visualization, Writing - original draft, Writing - review and editing. All authors reviewed and
398 approved the final manuscript.

399 **Conflicts of Interest Declaration**

400 All authors declare that they have no conflicts of interest.

401 **Funding**

402 This research received no specific grant from any funding agency in the public, commercial, or not-
403 for-profit sectors.

404 **Declaration of Generative AI and AI-Assisted Technologies in Manuscript Preparation**

405 During the preparation of this work, the authors used ChatGPT for language editing, manuscript
406 polishing, and formatting checks. After using this tool, the authors reviewed and edited the content as
407 needed and take full responsibility for the content of the publication.

408 **Data Availability**

409 Code, configurations, controlled raw results, summaries, paired statistics, NFE and initial-population
410 audits, figures and reproduction commands are archived in the local research repository. The existing
411 V2 result files remain byte-identical; all reported data are synthetic, and no proprietary, confidential or
412 human-subject data were used.

413

414 **References**

- 415 [1] Mao, Y., You, C., Zhang, J., Huang, K. and Letaief, K.B. (2017) 'A survey on mobile edge computing: the communication
416 perspective', IEEE Communications Surveys & Tutorials, Vol. 19, No. 4, pp.2322-2358, doi: 10.1109/COMST.2017.2745201.
417 [2] Taleb, T., Samdanis, K., Mada, B., Flinck, H., Dutta, S. and Sabella, D. (2017) 'On multi-access edge computing: a survey of the
418 emerging 5G network edge cloud architecture and orchestration', IEEE Communications Surveys & Tutorials, Vol. 19, No. 3,
419 pp.1657-1681, doi: 10.1109/COMST.2017.2705720.
420 [3] Sardellitti, S., Scutari, G. and Barbarossa, S. (2015) 'Joint optimization of radio and computational resources for multicell mobile-
421 edge computing', IEEE Transactions on Signal and Information Processing over Networks, Vol. 1, No. 2, pp.89-103, doi:
422 10.1109/TSIPN.2015.2448520.

- [4] Tang, Z., Sun, Z., Yang, N. and Zhou, X. (2023) 'Age of information of multi-user mobile edge computing systems', IEEE Open Journal of the Communications Society, Vol. 4, pp.1600-1614, doi: 10.1109/OJCOMS.2023.3294942.
- [5] Skorin-Kapov, L., Varela, M., Hossfeld, T. and Chen, K.-T. (2018) 'A survey of emerging concepts and challenges for QoE management of multimedia services', ACM Transactions on Multimedia Computing, Communications, and Applications, Vol. 14, No. 2s, pp.1-29, doi: 10.1145/3176648.
- [6] Luo, J., Deng, X., Zhang, H. and Qi, H. (2019) 'QoE-driven computation offloading for edge computing', Journal of Systems Architecture, Vol. 97, pp.34-39, doi: 10.1016/j.sysarc.2019.01.019.
- [7] Zhou, J. and Zhang, X. (2021) 'Fairness-aware task offloading and resource allocation in cooperative mobile-edge computing', IEEE Internet of Things Journal, Vol. 9, No. 5, pp.3812-3824, doi: 10.1109/IIOT.2021.3100253.
- [8] Dong, S., Tang, J., Abbas, K., Hou, R., Kamruzzaman, J., Rutkowski, L. and Buyya, R. (2024) 'Task offloading strategies for mobile edge computing: a survey', Computer Networks, Vol. 254, Article 110791, doi: 10.1016/j.comnet.2024.110791.
- [9] Wang, D., Bin Abu Bakar, K., Isyaku, B., Eisa, T.A.E. and Abdelmaboud, A. (2024) 'A comprehensive review on Internet of Things task offloading in multi-access edge computing', Heliyon, Vol. 10, No. 9, Article e29916, doi: 10.1016/j.heliyon.2024.e29916.
- [10] Wu, H., Lu, Y., Ma, H., Xing, L., Deng, K. and Lu, X. (2025) 'A survey on task type-based computation offloading in mobile edge networks', Ad Hoc Networks, Vol. 169, Article 103754, doi: 10.1016/j.adhoc.2025.103754.
- [11] Zhang, K., Mao, Y., Leng, S., Zhao, Q., Li, L., Peng, X., Pan, L., Maharjan, S. and Zhang, Y. (2016) 'Energy-efficient offloading for mobile edge computing in 5G heterogeneous networks', IEEE Access, Vol. 4, pp.5896-5907, doi: 10.1109/ACCESS.2016.2597169.
- [12] Almuselem, W. (2025) 'Deep reinforcement learning-enabled computation offloading: a novel framework to energy optimization and security-aware in vehicular edge-cloud computing networks', Sensors, Vol. 25, No. 7, Article 2039, doi: 10.3390/s25072039.
- [13] An, X., Li, Y., Chen, Y. and Li, T. (2024) 'Joint task offloading and resource allocation for multi-user collaborative mobile edge computing', Computer Networks, Vol. 250, Article 110604, doi: 10.1016/j.comnet.2024.110604.
- [14] Chen, T., Tan, F., Ai, J., Xiong, X., Wu, C. and Ren, X. (2024) 'Joint optimization of task offloading and service placement for digital twin empowered mobile edge computing', Proceedings of the 2024 3rd International Conference on Networks, Communications and Information Technology, pp.132-137, doi: 10.1145/3672121.3672145.
- [15] Huang, L., Bi, S. and Zhang, Y.-J.A. (2019) 'Deep reinforcement learning for online computation offloading in wireless powered mobile-edge computing networks', IEEE Transactions on Mobile Computing, Vol. 19, No. 11, pp.2581-2593, doi: 10.1109/TMC.2019.2928811.
- [16] Liu, L., Feng, J., Mu, X., Pei, Q., Lan, D. and Xiao, M. (2023) 'Asynchronous deep reinforcement learning for collaborative task computing and on-demand resource allocation in vehicular edge computing', IEEE Transactions on Intelligent Transportation Systems, Vol. 24, No. 12, pp.15513-15526, doi: 10.1109/TITS.2023.3249745.
- [17] Zang, L., Zhang, X. and Guo, B. (2022) 'Federated deep reinforcement learning for online task offloading and resource allocation in WPC-MEC networks', IEEE Access, Vol. 10, pp.9856-9867, doi: 10.1109/ACCESS.2022.3144415.
- [18] Abd-Elmagid, M.A., Pappas, N. and Dhillon, H.S. (2019) 'On the role of age of information in the Internet of Things', IEEE Communications Magazine, Vol. 57, No. 12, pp.72-77, doi: 10.1109/MCOM.001.1900041.
- [19] Jin, L., Tang, M., Pan, J., Zhang, M. and Wang, H. (2024) 'Asynchronous fractional multi-agent deep reinforcement learning for age-minimal mobile edge computing', arXiv:2409.16832.
- [20] Pei, Y., Zhao, Y. and Hou, F. (2024) 'Minimizing age of information in UAV-assisted edge computing system with multiple transmission modes', Tsinghua Science and Technology, Vol. 30, No. 3, pp.1060-1078, doi: 10.26599/TST.2024.9010046.
- [21] Bai, J., Su, H., Bi, J., Zhang, L., Yuan, X. and Liu, X. (2025) 'TLBO-HHO Aol-aware task offloading for mobile edge computing', Proceedings of the 2025 IEEE International Conference on Smart Internet of Things (SmartIoT), pp.88-95, doi: 10.1109/SmartIoT6867.2025.00021.
- [22] Fiedler, M., Hossfeld, T. and Tran-Gia, P. (2010) 'A generic quantitative relationship between quality of experience and quality of service', IEEE Network, Vol. 24, No. 2, pp.36-41, doi: 10.1109/MNET.2010.5430142.
- [23] Jain, R.K., Chiu, D.M. and Hawe, W.R. (1984) 'A quantitative measure of fairness and discrimination for resource allocation in shared computer systems', Digital Equipment Corporation, Hudson, MA, Technical Report DEC-TR-301.
- [24] Kelly, F.P., Maulloo, A.K. and Tan, D.K.H. (1998) 'Rate control for communication networks: shadow prices, proportional fairness and stability', Journal of the Operational Research Society, Vol. 49, No. 3, pp.237-252, doi: 10.1038/sj.jors.2600523.
- [25] Le Callet, P., Moller, S. and Perkis, A. (eds) (2013) 'Qualinet white paper on definitions of quality of experience', European Network on Quality of Experience in Multimedia Systems and Services.
- [26] Ling, C., Zhang, W., He, H., Yadav, R., Wang, J. and Wang, D. (2024) 'QoS and fairness oriented dynamic computation offloading in the Internet of Vehicles based on estimated time of arrival', IEEE Transactions on Vehicular Technology, Vol. 73, No. 7, pp.10554-10571, doi: 10.1109/TVT.2024.3364669.
- [27] Chen, H., Heidari, A.A., Chen, H., Wang, M., Pan, Z. and Gandomi, A.H. (2020) 'Multi-population differential evolution-assisted Harris hawks optimization: framework and case studies', Future Generation Computer Systems, Vol. 111, pp.175-198, doi: 10.1016/j.future.2020.04.008.
- [28] Deb, K., Pratap, A., Agarwal, S. and Meyarivan, T. (2002) 'A fast and elitist multiobjective genetic algorithm: NSGA-II', IEEE Transactions on Evolutionary Computation, Vol. 6, No. 2, pp.182-197, doi: 10.1109/4235.996017.
- [29] Deb, K., Sindhya, K. and Hakanen, J. (2016) 'Multi-objective optimization', in Decision Sciences: Theory and Practice, CRC Press, Boca Raton, FL, pp.161-200.
- [30] Heidari, A.A., Mirjalili, S., Faris, H., Aljarah, I., Mafarja, M. and Chen, H. (2019) 'Harris hawks optimization: algorithm and applications', Future Generation Computer Systems, Vol. 97, pp.849-872, doi: 10.1016/j.future.2019.02.028.
- [31] Li, Z. and Zhu, Q. (2020) 'Genetic algorithm-based optimization of offloading and resource allocation in mobile-edge computing', Information, Vol. 11, No. 2, Article 83, doi: 10.3390/info11020083.
- [32] Rao, R.V., Savsani, V.J. and Vakharia, D.P. (2011) 'Teaching-learning-based optimization: a novel method for constrained mechanical design optimization problems', Computer-Aided Design, Vol. 43, No. 3, pp.303-315, doi: 10.1016/j.cad.2010.12.015.
- [33] Zhang, Q. and Li, H. (2007) 'MOEA/D: a multiobjective evolutionary algorithm based on decomposition', IEEE Transactions on Evolutionary Computation, Vol. 11, No. 6, pp.712-731, doi: 10.1109/TEVC.2007.892759.
- [34] Su, H., Zhao, D., Heidari, A.A., Liu, L., Zhang, X., Mafarja, M. and Chen, H. (2023) 'RIME: a physics-based optimization', Neurocomputing, Vol. 532, pp.183-214, doi: 10.1016/j.neucom.2023.02.010.
- [35] Xue, J. and Shen, B. (2023) 'Dung beetle optimizer: a new meta-heuristic algorithm for global optimization', The Journal of Supercomputing, Vol. 79, No. 7, pp.7305-7336, doi: 10.1007/s11227-022-04959-6.

- 495 [36] Yang, X., Liu, J., Liu, Y., Xu, P., Yu, L., Zhu, L., Chen, H. and Deng, W. (2021) 'A novel adaptive sparrow search algorithm based
496 on chaotic mapping and t-distribution mutation', *Applied Sciences*, Vol. 11, No. 23, Article 11192, doi: 10.3390/app112311192.
- 497 [37] Ma, C., Zhu, J., Liu, M., Zhao, H., Liu, N. and Zou, X. (2021) 'Parking edge computing: parked-
498 vehicle-assisted task offloading for urban VANETs', *IEEE Internet of Things Journal*, Vol. 8, No.
499 11, pp.9344-9358, doi: 10.1109/JIOT.2021.3056396.
- 500 [38] Wu, H., Wolter, K., Jiao, P., Deng, Y., Zhao, Y. and Xu, M. (2021) 'EEDTO: an energy-efficient
501 dynamic task offloading algorithm for blockchain-enabled IoT-edge-cloud orchestrated
502 computing', *IEEE Internet of Things Journal*, Vol. 8, No. 4, pp.2163-2176, doi:
503 10.1109/JIOT.2020.3033521.
- 504 [39] Feng, J., Yu, F.R., Pei, Q., Chu, X., Du, J. and Zhu, L. (2020) 'Cooperative computation
505 offloading and resource allocation for blockchain-enabled mobile-edge computing: a deep
506 reinforcement learning approach', *IEEE Internet of Things Journal*, Vol. 7, No. 7, pp.6214-6228,
507 doi: 10.1109/JIOT.2019.2961707.
- 508 [40] Yuan, X., Tian, H., Wang, H., Su, H., Liu, J. and Taherkordi, A. (2020) 'Edge-enabled WBANs
509 for efficient QoS provisioning healthcare monitoring: a two-stage potential game-based
510 computation offloading strategy', *IEEE Access*, Vol. 8, pp.92718-92730, doi:
511 10.1109/ACCESS.2020.2992639.